Huawei Access Node BAS-ACC-05 QoS Congestion Event Resolution SOP

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3943

Description

This Standard Operating Procedure (SOP) outlines the corrective actions for resolving QoS/Congestion Events on Huawei Access Node BAS-ACC-05, located at Basavanagudi POP. The failure domain in question is triggered by Microbursts, interface tail drops, and QoS queue starvation. These occurrences are a direct consequence of the Huawei Access Node's (BAS-ACC-05) inability to handle the surge in traffic, leading to an overload situation where the network's Quality of Service (QoS) parameters are being compromised. The root cause of this issue is primarily attributed to the Huawei Access Node's (BAS-ACC-05) limitations in handling high-speed packet processing, coupled with inadequate configuration and tuning of the Quality of Service (QoS) mechanisms. This is further exacerbated by the lack of adequate hardware resources (e.g., CPU, memory) to handle the increased traffic load, leading to a QoS/Congestion Event.

Pre-requisite Checks

- check: Verify that the redundant link is up and operational.

- check: Ensure that the BAS-ACC-05 node is configured with adequate CPU, memory, and other hardware resources to handle the expected traffic load.

- check: Verify that the Quality of Service (QoS) configuration is adequate and properly tuned to handle the expected traffic patterns.

Potential Root Causes

Cause	Probability
Inadequate Quality of Service (QoS) configuration.	High
Hardware resource limitations (CPU, memory).	Medium
Microbursts and interface tail drops.	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-queue-info, Expected Output: Queue
information for interface Eth1/0/0:, Description: This command is used
to verify the current queue information for the interface Eth1/0/0,
which is experiencing congestion. The expected output will indicate
the number of packets in each queue, which will help in identifying
the root cause of the congestion.

Command: display interface Eth1/0/0 queue statistics, Expected Output:
Queue statistics for interface Eth1/0/0:, Description: This command is
used to verify the queue statistics for the interface Eth1/0/0, which
```

```
is experiencing congestion. The expected output will indicate the
number of packets dropped or discarded due to queue congestion.
```

Resolution Steps

Step 1: Step: Adjust the Quality of Service (QoS) configuration, Command: set qos profile profile1 queue1 threshold 10000, Description: This command is used to adjust the Quality of Service (QoS) configuration for the interface Eth1/0/0 by setting the queue threshold to 10000 packets. This will help in preventing queue congestion and ensuring that packets are not dropped or discarded due to queue overflow.

Step 2: Step: Increase hardware resources (CPU, memory), Command: upgrade cpu 2G, Description: This command is used to increase the CPU resources for the BAS-ACC-05 node from 1G to 2G. This will help in handling the increased traffic load and preventing queue congestion.

Post-Check Verification

```
Command: display current-queue-info, Expected Output: Queue
information for interface Eth1/0/0:, Description: This command is used
to verify that the Quality of Service (QoS) configuration has been
successfully adjusted and that the queue information is within
acceptable limits.

Command: display interface Eth1/0/0 queue statistics, Expected Output:
Queue statistics for interface Eth1/0/0:, Description: This command is
used to verify that the queue statistics are within acceptable limits
and that packets are not being dropped or discarded due to queue
congestion.
```

Roll-back Procedures

- Step: Restore original Quality of Service (QoS) configuration, Command: restore qos profile profile1 queue1 threshold 5000, Description: This command is used to restore the original Quality of Service (QoS) configuration for the interface Eth1/0/0. This will help in reverting to the previous state and preventing any further disruptions to the network.

- Step: Downgrade hardware resources (CPU), Command: downgrade cpu 1G, Description: This command is used to downgrade the CPU resources for the BAS-ACC-05 node from 2G to 1G. This will help in reverting to the previous state and preventing any further disruptions to the network.